MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OFAERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**AUTOMOBILE ENGINEERING**	**Course Type**	**C**
Course Name	**BASICS OF I C ENGINES**	**Course Code**	**AAE 2129**
Semester	**III**	**Max Marks**	**50**
Date	**29-11-2024**	**Duration**	**3 hours**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	An SI engine has a fuel air ratio of 0.07:1. How many kilograms of air per hour are required for an output of 75 kW at an overall efficiency of 20%? How many m^3 of air are required per hour if the density of the air is 1.2 kg/m^3? If the fuel vapour has a density four times that of air, how many m^3 per hour of the mixture is required? The calorfic value of the fuel is 43700 kJ/kg.	**04**	CO4	1, 2, 3, 4, 5,10, 12	**4**	1,2	**4**
1B	Explain the phenomenon and stages of flame propagation in a SI engine combustion process.	**03**	CO2	1, 2, 3, 4, 5,10, 12	**2**	1,2,13	**2**
1C	What are F head combustion chambers? With example, discuss the two important features of the F head designs.	**03**	CO2	1, 2, 3, 4, 5,10, 12	**2**	1,2,13	**3**
2A	Explain the term delay period in CI engine. Identify the stages involved in delay period and expalian the importance of each stage in CI engine combustion process.	**03**	CO2	1, 2, 3, 4, 5,10, 12	**2**	1,2,13	**3**
2B	A four-cylinder, four-stroke diesel engine develops a power of 180 kW at 1500 rpm. The bsfc is 0.2 kg/kW-h. At the beginning of injection, pressure is 30 bar and the maximum cylinder pressure is 50 bar. The injection is	**04**	CO2	1, 2, 3, 4, 5,10, 12	**2**	1,2,13	**4**

	expected to be at 200 bar and maximum pressure at the injector is set to be about 500 bar. Assuming the following: • C_d for injector = 0.7 • S.G. of fuel = 0.875 • Atmospheric pressure = 1 bar • Effective pressure difference = Average pressure difference over the injection period. Determine the total orifice area required per injector if the injection takes place over 15° crank angles.						
2C	Ellobrate the working and features of a Acceleration pump system of a SI engine carburettor.	**03**	CO2	1, 2, 3, 4, 5,10, 12	**2**	1,2,13	**2**
3A	Describe any two types of injection nozzles and discuss their relative advantages and disadvantages.	**04**	CO2	1, 2, 3, 4, 5,10, 12	**2**	1,2,13	**2**
3B	Explain the factors which must be considered before deciding the firing order of a multicylinder engine.Also list typical firing order of a 3, 4 and 6 cylinder in line IC engine.	**03**	CO3	1, 2, 3, 4, 5,10, 12	**3**	1,2	**3**
3C	Explain the effect of following parameters on engine friction. a) Compression ratio, b) Cooling water temperature, c) Oil viscosity	**03**	CO3	1, 2, 3, 4, 5,10, 12	**3**	1,2	**3**
4A	With neat layout explain the working of a wet sump lubrication system.	**03**	CO3	1, 2, 3, 4, 5,10, 12	**3**	1,2	**2**
4B	Describe with a neat sketch the forced circulation system and list the advantages of the same.	**04**	CO3	1, 2, 3, 4, 5,10, 12	**3**	1,2	**2**
4C	Discuss the design features of any two types of radiator matrices commonly used. What are their relative advantages and disadvantages.	**03**	CO3	1, 2, 3, 4, 5,10, 12	**3**	1,2	**2**
5A	List the various methods for measuring the frictional power of IC Engine. Describe the Morse test method of measurement of frictional power.	**02**	CO4	1, 2, 3, 4, 5,10, 12	**4**	1,2	**2**

5B	The following data were recorded from a test on a single cylinder four stroke oil engine. Cylinder bore=150 mm, engine stroke=250 mm, area of indicator diagram=450 mm^2, length of indicator diagram=50 mm, indicator spring rating=1.2 bar/mm, engine speed=420 rpm, brake torque=217 Nm, fuel consumption= 2.95 kg/hr, calorific value of fuel=44000 kJ/kg, cooling water rate of flow=0.068 kg/sec, cooling water temperature rise=45 K, specific heat capacity of the cooling water= 4.1868 kJ/kg K. On test, the engine is enclosed in a thermally and acoustically insulated box, through which the output drive, water, fuel, air and exhaust connections pass. Ventilating air blown up through the box at the rate of 14 kg/min enters at 10°C and leaves at 55°C. Take specific heat of dry exhaust as 1.005 kJ/kg. Calculate mechanical efficiency, brake thermal efficiency, specific fuel consumption and draw up an energy balance in kW.	**05**	CO4	1, 2, 3, 4, 5,10, 12	**4**	1,2	**4**
5C	With neat layout explain the process of two stage turbocharging and discuss its advantages and disadvantages.	**03**	CO1	1, 2, 3, 4, 5,10, 12	**1**	1,12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	**03**
2	**20**
3	**16**
4	**11**
5	

BL	Total Marks
2	**25**
3	**12**
4	**13**
5	